Assignment9 – Learning

Given: June 17 Due: June 22

Problem 9.1 (Logical Formulation of Learning)

Some people with different attributes go sunbathing, the result class is whether they get sunburned:

#	Hair	Height	Weight	Lotion	Sunburned
1	Blonde	Short	Light	No	Yes
2	Blonde	Short	Average	Yes	No
3	Brown	Short	Average	Yes	No
4	Blonde	Short	Average	No	Yes
5	Blonde	Tall	Heavy	No	Yes
6	Brown	Tall	Heavy	No	No

We want to model these examples in first-order logic.

1. Explain which predicates are needed. For each predicate give the arity and the domain of each argument. Give the formal representation of the description and the classification of Example 1.

Solution: We use a unary predicate for each boolean attribute/class and a binary predicate for the others. The first argument of all predicates represents the example and has domain $\{1, \dots, 6\}$. The domain of the second argument of the binary predicates is the set of possible values. Thus, we have for an example e:

- $Hair(e, c)$ for $c \in \{Blonde, Brown\}$
- $Height(e, h)$ for $h \in \{Short, Tall\}$
- $Weight(e, w)$ for $w \in \{Light, Average, Heavy\}$
- $Lotion(e)$
- $Sunburned(e)$

For exampel, the description of example 1 is $Hair(1, Blonde) \wedge Height(1, Short) \wedge Weight(1, Light) \wedge \neg Lotion(1)$. Its classification is $Sunburned(1)$.

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2. For attributes A, B, we write $A_1, \dots, A_n \succ B$ if any two examples that agree on all of the attributes A_i also agree on the attribute B.
Explain whether Height$\succ$ Weight and Weight $\succ$ Height hold or do not hold.

Solution: Height $\succ$ Weight does not hold: Examples 1 and 2 agree on Height but not on Weight. Weight $\succ$ Height holds: any pair of examples that agree on Weight also agree on Height.

If two people have the same values for all attributes in $\mathcal{A}$, then they must also have the same value for Sunburned.

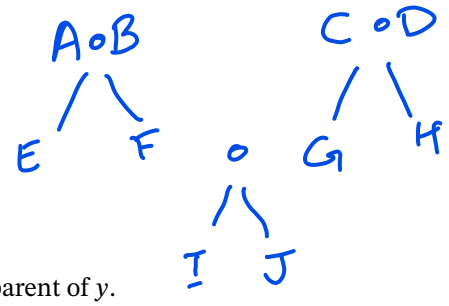
For all individuals e and e′: If e and e′ have the same Hair color and the same Lotion status, then they must have the same Sunburned value

3. Give a minimal subset $\mathcal{A} \subseteq \{Hair, Height, Weight, Lotion\}$ such that $\mathcal{A} \succ Sunburned$ holds.
Using the predicates introduced above, give a logical formula that captures the rule $\mathcal{A} \succ Sunburned$.

Solution: $\mathcal{A} = \{Hair, Lotion\}$. This corresponds to the formula $\forall e, e'.\forall c.\big((Hair(e,c) \wedge Hair(e',c)) \wedge ((Lotion(e) \Leftrightarrow Lotion(e'))) \Rightarrow ((Sunburned(e) \Leftrightarrow Sunburned(e'))\big)$.
For example, instantiating with $e' = 1$ and $c = Blonde$, that implies $\forall e.Hair(e, Blonde) \wedge \neg Lotion(e) \Rightarrow Sunburned(e)$.

Problem 9.2 (Inductive Learning)

Consider the family tree given by the following relations:



couple	children
A, B	E, F
C, D	G, H
F, G	I, J

Assume we already know the predicate $\texttt{par}(x, y)$ for x being a parent of y.

Our goal is to learn the predicate $\texttt{gp}(x, y)$ for x being a grandparent of y. That means to find a formula D such that $\forall x, y.\texttt{gp}(x, y) \Leftrightarrow D(x, y)$.

We do not know D, but we have the following examples for gp:

person-pair	grandparent
A, I	yes
B, I	yes
A, J	yes
A, E	no
A, F	no
F, A	no
A, A	no
C, J	yes
D, H	no
I, A	no

a is the grandparent of I

1. Give the intended formula D_1, i.e., the correct definition of grandparent.

Solution: $D_1(x, y) = \exists u.\texttt{par}(x, u) \wedge \texttt{par}(u, y)$

There exists some person u such that x is a parent of u and u is a parent of y.

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2. Give a formula D_2 that covers exactly the positive examples.

Solution: $D_2(x, y) = (x = A \wedge y = I) \vee (x = B \wedge y = I) \vee$ and so on for all

$\vee (x = A \wedge y = J)$
$\vee (x = C \wedge y = J)$

positive examples.

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build D2 by just OR-ing together equalities for each positive example.
That's straightforward and takes $O(n)$ time for n examples

- hard to learn
- doesnt generalize- if dataset is incomplete b2 iwll be wrong for uneseen pairs
- d2 grows w data the forumla gets longer and longer

3. Explain the pros and cons of learning the formula D as D_1 vs. D_2.

- D1 works for all current and future people in the family tree, not just the ones in the example table. If a new grandchild appears, D1 will classify it correctly without changing the formula.

- constant size doesnt change w no of exmp

Solution: pros of D_1:

- D_2 captures only the provided examples. If that list is incomplete, it is likely the wrong formula. Even if the list is complete, D would have to be changed every time the list of examples changes. D_1 captures all future examples correctly.
- D_1 has size $O(1)$, whereas D_2 has size $O(n)$ where n is the number of examples.

pros of D_2:

- D_2 can be easily read off the list of examples in $O(n)$ time.
- A nice formula for D_1 might not even exist or might be very difficult to find or might be impossible to find from the given examples.

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4. We want to learn algorithmically the formula $D(x, y) = \exists u_1, \ldots, u_l . L_1 \wedge \ldots \wedge L_k$ where each L_i is a literal of the form $P(x, y, u_1, \ldots, u_l)$ or $\neg P(x, y, u_1, \ldots, u_l)$ for some predicate symbol P including the equality predicate, i.e., $P \in \{\texttt{par}, \texttt{gp}, =\}$. We do so by building the set $\{L_1, \ldots, L_k\}$ of literals gradually (with the understanding that each L_i can have any free variables in addition to x and y, which we will collect at the end as the u_i).
   1. If we start with the empty set of literals, give all useful choices that we can make for the first literal.
   2. For each choice L that is non-recursive (i.e., P is not the target predicate gp), positive (i.e., the literal is not negated), not an equality (i.e., P is not the equality predicate $=$), and does not introduce a new variable (i.e., only uses x and y), which examples are falsely classified?

variables x y u v

A literal is either a predicate or its negation. P(x) , not P(x)

predicates p being
par
gp
=

Solution:

8 literals without new variables
only x, y combos

8 literals with one new variable
u so x and y and u binary combos

1. There are 16 choices each for $P \in \{\texttt{par}, \texttt{gp}\}$: $P(x, x)$, $P(y, y)$ $P(x, y)$ or $P(y, x)$ (no new variables), $P(x, u)$, $P(u, x)$, $P(y, u)$, $P(u, y)$ (1 new variable); as well as the negated versions. (It is useless to add $P(u, v)$ or $P(u, u)$ where all variables are new.)
   Additionally, we can choose $x = y$ and $\neg x = y$. (It is useless to add an (in)equality with a new variable or where both variables are the same.)
2. There are 4 such choices:
   - $\texttt{par}(x, x)$: no false-positives
   - $\texttt{par}(y, y)$: no false-positives
   - $\texttt{par}(x, y)$: false-positives are (A, E), (A, F), and (D, H)
   - $\texttt{par}(y, x)$: false-positives are (F, A)

   Additionally, for all 4 choices, all positive examples are false-negatives.

USEFUL - help classify some pairs as positive and others as negative. ie. The literal depend on x or y in a way that can change the classification of different (x,y) examples.

Non-recursive: only par, not gp.

Positive: only par(...), not ¬par(...).

Not equality: ignore x = y, ¬(x = y).

No new variables: only use x and y, no u.

Problem 9.3 (De/Ab/Induction)

1. Explain the differences between deduction, abduction, and induction.

Solution: Deduction finds facts that are a consequence of known facts. For example, if we know $A \Rightarrow B$ and A, we can infer B. Sound deduction rules guarantee that the inferred facts are true if the hypotheses are.
Abduction is not a form of logical consequence and occasionally even constitutes faulty reasoning.* For example, it conjectures A from knowing $A \Rightarrow B$ and B. Abduction rules can be seen as learning likely explanations for given observations.
Induction finds general rules by abstracting from more specific facts. For example, knowing $A(1)$ and $A(2)$, we can abstract $\forall x.A(x)$. This can be very helpful to learn rules from observations. Induction can lead to over-generalization if not enough examples are present. If the examples cover all possible cases, it is valid inference principle (as induction on the natural numbers).

$$\frac{A \Rightarrow B \quad [A]}{B}\,D$$

$$\frac{A \Rightarrow B \quad B}{A}\,A$$

* Remember:
F ⇒ T √√
T ⇒ F ✗
F ⇒ F √√
T ⇒ T √√

$$\frac{A(1) \quad A(2)}{\forall x.A(x)}\,I$$